

Inverse Trigonometric Functions: Formula Sheet

Class 12 Mathematics – Chapter 2 Complete Reference & Formulas

1 Principal Value Branches (Domains and Ranges)

Function	Domain	Principal Value Branch (Range)
$y = \sin^{-1} x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$y = \cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$y = \operatorname{cosec}^{-1} x$	$\mathbb{R} - (-1, 1)$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$
$y = \sec^{-1} x$	$\mathbb{R} - (-1, 1)$	$[0, \pi] - \{\frac{\pi}{2}\}$
$y = \tan^{-1} x$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$y = \cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$

2 Fundamental Inverse-Function Properties

- Cancellation / Inversion Identities:

$\sin(\sin^{-1} x) = x, \quad x \in [-1, 1]$	$\sin^{-1}(\sin x) = x, \quad x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos(\cos^{-1} x) = x, \quad x \in [-1, 1]$	$\cos^{-1}(\cos x) = x, \quad x \in [0, \pi]$
$\tan(\tan^{-1} x) = x, \quad x \in \mathbb{R}$	$\tan^{-1}(\tan x) = x, \quad x \in (-\frac{\pi}{2}, \frac{\pi}{2})$
$\operatorname{cosec}(\operatorname{cosec}^{-1} x) = x, \quad x \geq 1$	$\operatorname{cosec}^{-1}(\operatorname{cosec} x) = x, \quad x \in [-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$
$\sec(\sec^{-1} x) = x, \quad x \geq 1$	$\sec^{-1}(\sec x) = x, \quad x \in [0, \pi] - \{\frac{\pi}{2}\}$
$\cot(\cot^{-1} x) = x, \quad x \in \mathbb{R}$	$\cot^{-1}(\cot x) = x, \quad x \in (0, \pi)$

3 Reciprocal & Negative Argument Identities

- **Reciprocal Relations:**

$$\sin^{-1}\left(\frac{1}{x}\right) = \operatorname{cosec}^{-1}x, \quad x \geq 1 \text{ or } x \leq -1$$

$$\cos^{-1}\left(\frac{1}{x}\right) = \sec^{-1}x, \quad x \geq 1 \text{ or } x \leq -1$$

$$\tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1}x, \quad x > 0$$

- **Negative Argument Properties:**

$$\sin^{-1}(-x) = -\sin^{-1}x, \quad x \in [-1, 1]$$

$$\tan^{-1}(-x) = -\tan^{-1}x, \quad x \in \mathbb{R}$$

$$\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}x, \quad |x| \geq 1$$

$$\cos^{-1}(-x) = \pi - \cos^{-1}x, \quad x \in [-1, 1]$$

$$\sec^{-1}(-x) = \pi - \sec^{-1}x, \quad |x| \geq 1$$

$$\cot^{-1}(-x) = \pi - \cot^{-1}x, \quad x \in \mathbb{R}$$

4 Complementary Angle & Addition Formulas

- **Complementary Identities:**

$$\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}, \quad x \in [-1, 1]$$

$$\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}, \quad x \in \mathbb{R}$$

$$\sec^{-1}x + \operatorname{cosec}^{-1}x = \frac{\pi}{2}, \quad |x| \geq 1$$

- **Addition and Subtraction of Inverse Tangents:**

$$\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right), \quad xy < 1$$

$$\tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\frac{x-y}{1+xy}\right), \quad xy > -1$$

5 Multiple Angle Identities & Transformations

- **Double Angle Formulas for Inverse Tangent ($2 \tan^{-1} x$):**

$$2 \tan^{-1} x = \sin^{-1} \left(\frac{2x}{1+x^2} \right), \quad |x| \leq 1$$

$$2 \tan^{-1} x = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right), \quad x \geq 0$$

$$2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1-x^2} \right), \quad -1 < x < 1$$

- **Double and Triple Angle Formulas for Inverse Sine & Cosine:**

$$\sin^{-1} \left(2x\sqrt{1-x^2} \right) = \begin{cases} 2 \sin^{-1} x, & -\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}} \\ 2 \cos^{-1} x, & \frac{1}{\sqrt{2}} \leq x \leq 1 \end{cases}$$

$$3 \sin^{-1} x = \sin^{-1} (3x - 4x^3), \quad x \in \left[-\frac{1}{2}, \frac{1}{2} \right]$$

$$3 \cos^{-1} x = \cos^{-1} (4x^3 - 3x), \quad x \in \left[\frac{1}{2}, 1 \right]$$

$$3 \tan^{-1} x = \tan^{-1} \left(\frac{3x - x^3}{1 - 3x^2} \right), \quad |x| < \frac{1}{\sqrt{3}}$$